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Low-Reynolds-number Airfoil Design Optimization using Deep-learning-based Tailored Airfoil Modes

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Low-Reynolds-number high-lift airfoil design is critical to the performance of unmanned aerial vehicles (UAV). However, since laminar-to-turbulent transition dominates the aerodynamic performance of low-Reynolds-number airfoils and the transition position may exhibit an abrupt change even with a small geometric deformation, aerodynamic coefficient functions become discontinuous in this regime, which brings significant difficulties to the application of conventional aerodynamic design optimization methods. To efficiently perform low-Reynolds-number airfoil design, we present a tailored airfoil modal parameterization method, which reasonably defines the desired design space using deep-learning techniques. Coupled with surrogate-based optimization, the proposed method has shown to be effective and efficient in low-Reynolds-number high-lift airfoil design. It is found that it is necessary to consider laminar-to-turbulent transition and to perform multi-point optimization in practical low-Reynolds-number airfoil design. The maximal lift coefficient is an active constraint influencing the selection of the optimal cruise lift coefficient. The results show the complexity of low-Reynolds-number high-lift airfoil design and highlight the significance of the proposed method in the improvement of optimization efficiency.

I. Introduction

UAVs have been widely used in various fields [1]. The aerodynamic shape, especially the airfoil shape, is critical to the performance of UAVs. With diverse design demands and short design cycles of UAVs, it is of great importance to rapidly develop the optimal aerodynamic shape [2, 3]. This poses higher requirements on the efficiency of airfoil shape design optimization [4].

Aerodynamic shape optimization using numerical simulations can automatically find the optimal design that maximizes the aerodynamic performance. Popular aerodynamic optimization methods are composed of two categories: simulation-based optimization and surrogate-based optimization. Simulation-based optimization directly couples an optimizer with a numerical simulation solver, so the objective function evaluation is computationally expensive, which puts a high-efficiency demand on the optimizer. High-dimensional design variables are generally required in aerodynamic shape optimization, which make gradient-free optimization algorithms extremely inefficient. Thus, a gradient-based optimizer is almost necessary for simulation-based optimization [5, 6], and the aerodynamic derivatives are usually solved by the adjoint method [7–10] in this circumstance. A potential concern for this method is the poor global searching capability. On the other hand, surrogate-based optimization [11–13] replaces costly numerical

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simulations with computational cheap approximation models in aerodynamic analysis, so gradient-free or multi-start gradient-based algorithms can be used to reduce the possibility of being stuck to local minima. This method is generally infill-based with the surrogate model being iteratively refined, and the selection of infill criteria is seeking for a balance of exploration and exploitation [12], such as the expected improvement (EI) criterion used in the efficient global optimization method [14]. Both categories of methods have been shown to be effective in aerodynamic shape design [15–20] especially in the transonic regime [21–25].

Nevertheless, there remain difficulties in low-Reynolds-number airfoil shape design optimization [26–29] due to the nonlinear and sensitive flow phenomena. Using the term “low-Reynolds-number”, we refer to a flow regime with the chord Reynolds number $Re < 1.0 \times 10^6$, and most UAVs operate in this regime with $Re \in [1.0 \times 10^5, 1.0 \times 10^6]$ [30–32]. The flow past the UAV airfoil usually encounters laminar-to-turbulent transition [33] in this regime, and thus the change of aerodynamic functions can be significant and abrupt [34] since the friction force in turbulent flows is much larger than that in laminar flows. We investigate the change of drag coefficient between two random airfoils with $Re = 3.8 \times 10^5$ and $\alpha = 3.0^\circ$. As shown in Fig. 1, with the change of the airfoil shape, the function of the drag coefficient is discontinuous due to abrupt moving-forward of the transition point at the lower surface of the airfoil. This means that the laminar-to-turbulent transition position is sensitive to even a small shape deformation. The influences of this issue are two-fold, which introduce challenges to both simulation-based optimization and surrogate-based optimization. On the one hand, the discontinuous objective function makes simulation-based optimization with gradient-based algorithms become unsuitable unless artificial smoothing measures are taken to filter out the discontinuities [35]. On the other hand, with the aerodynamic performance sensitive to small shape deformations, a detailed geometric parameterization with high-dimensional shape design variables is required to fully tap the potential of aerodynamic shape design. Surrogate-based optimization does not scale well with dimensionality and is inefficient in this circumstance. Thus, there is a strong demand to develop an optimization method that is efficient and effective in low-Reynolds-number airfoil design.

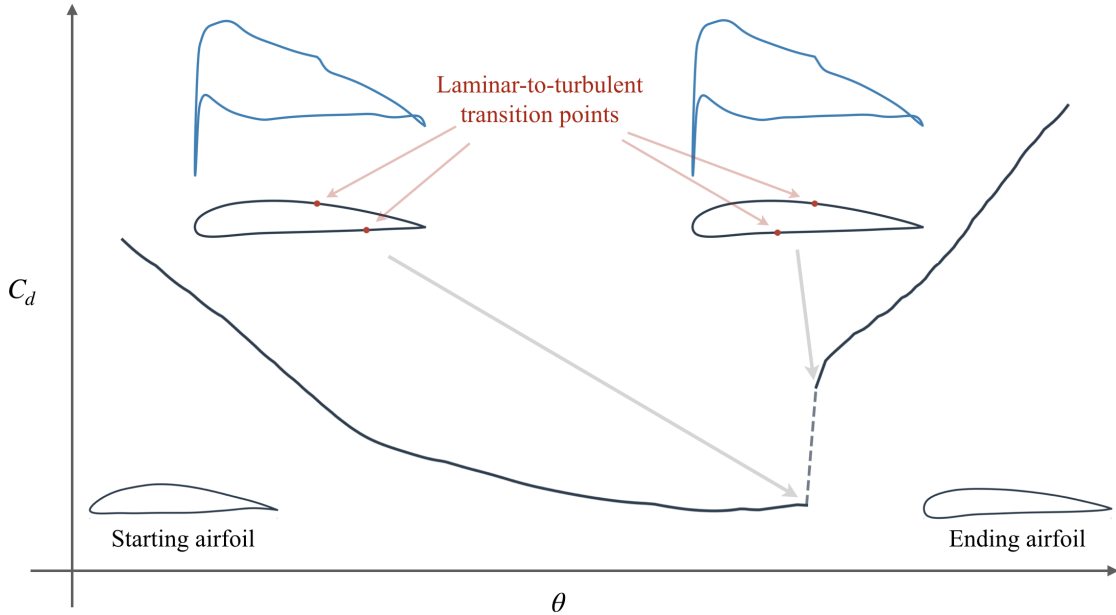


Fig. 1 In the low-Reynolds-number flow regime ($Re < 1.0 \times 10^6$), aerodynamic functions may be discontinuous due to the abrupt change of the laminar-to-turbulent transition position, which brings difficulties to design optimization. The intermediate airfoil shapes are obtained via linear interpolation between the starting and ending airfoils concerning a non-dimensional parameter θ .

To address the issue, we present an efficient surrogate-based airfoil design optimization method using a tailored modal parameterization based on deep-learning methods. The tailored airfoil modes are derived from customized airfoils satisfying geometric constraints of the design optimization problem, so they can have a stronger ability to parameterize the potential optimal airfoil shape. The customized airfoils are generated by a deep-learning-based optimal sampling method, where the realistic airfoil shapes are ensured by imposing a geometric validity constraint. Compared

with conventional airfoil modes derived from a general airfoil database, using the tailored airfoil modes is promising to define a compact design space with no geometric abnormality issues and improve optimization efficiency. To highlight this point, we couple the tailored modes with a surrogate-based optimization framework and investigate optimization efficiency in various low-Reynolds-number airfoil design cases. In addition, since single-point optimization may overstate the on-design performance, we investigate different multi-point optimization strategies to present an efficient approach to practical low-Reynolds-number airfoil design. As the maximal lift coefficient ($C_l^{\max}$) is a critical constraint in practical airfoil design, we investigate its influence on the optimal airfoil shape and the selection of the cruise C_l .

The rest of this paper is organized as follows. First, we describe the tailored modal parameterization method and optimization framework in Section II. Then, in Section III, we highlight the optimization efficiency by performing airfoil design using both conventional and tailored airfoil modes. In Section IV, we perform multi-point design optimization and investigate the influence of $C_l^{\max}$ in practical low-Reynolds-number high-lift airfoil design. Finally, in Section V, we end this paper with a summary of conclusions.

II. Methodology

A. Conventional Modal Parameterization with a Publicly-available Airfoil Database

Aerodynamic shape parameterization defines the geometric design space, and the choice of the parameterization method has a significant influence on aerodynamic design optimization. If the parameterization has limited geometric freedom, the optimal design cannot be found no matter which optimization method is used. If the parameterization defines a too large design space, optimization efficiency would be significantly reduced.

The modal parameterization using mode shapes [36, 37] has shown to be the most efficient method among several commonly-used airfoil parameterization approaches [38]. Airfoil mode shapes can be extracted from an airfoil database using the singular value decomposition (SVD), so they are usually called SVD modes. Assume that we have a database composed of m airfoils and each airfoil is represented by n surface points with the same x coordinate template. Then, with the average y value of the i th coordinate denoted as $\bar{y}_i$, the m airfoils are assembled into a snapshot matrix

$$\mathbf{A} = \begin{bmatrix} y_1^{(1)} - \bar{y}_1 & y_1^{(2)} - \bar{y}_1 & \dots & y_1^{(m)} - \bar{y}_1 \\ y_2^{(1)} - \bar{y}_2 & y_2^{(2)} - \bar{y}_2 & \dots & y_2^{(m)} - \bar{y}_2 \\ \vdots & \vdots & \ddots & \vdots \\ y_n^{(1)} - \bar{y}_n & y_n^{(2)} - \bar{y}_n & \dots & y_n^{(m)} - \bar{y}_n \end{bmatrix}. \quad (1)$$

Performing SVD on $\mathbf{A}$, we have

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T, \quad (2)$$

where the orthogonal columns in $\mathbf{U}$ are the y coordinates of the airfoil mode shapes. Then, an arbitrary airfoil $\mathbf{y}$ can be expressed as $\mathbf{y} = \mathbf{U}\mathbf{c} + \bar{\mathbf{y}}$, and $\mathbf{c}$ (mode coefficients) are the design variables of the modal parameterization for airfoil shape design.

As the largest publicly available airfoil database, the UIUC airfoil database is usually used to extract airfoil mode shapes. As explained in [37–39], a preprocessing step is generally required to smooth and formalize the UIUC airfoils. To this end, we use the open-source Python script provided by Li et al. [40]*. After the preprocessing, we have $m = 1407$ airfoils and each airfoil is represented by $n = 251$ surface points (x_i, y_i) , $i = 1, \dots, n$. Some of these airfoils are shown in Fig. 2, and the derived airfoil mode shapes are presented by showing the deformation effect on the same baseline airfoil.

In airfoil design optimization, the bounds of each mode coefficient can be defined by computing the minimum and maximum values of the corresponding coefficients in UIUC airfoils. To reduce the influence of a niche airfoil, we use the percentiles of 0.1% and 99.9% of UIUC airfoil mode coefficients to define the lower and upper bounds, respectively. As scanning of the design space, some sample airfoils generated by the Latin hypercube sampling (LHS) method are shown in Fig. 2. We find that most of the sample airfoils have abnormal shapes, which implies that the conventional airfoil modes define a much larger space than required. The geometric abnormality issue would bring unnecessary difficulties to design optimization since it is difficult to evaluate the performance of abnormal airfoils due to the non-convergence of aerodynamic analyses.

*https://github.com/npuljc/Airfoil_preprocessing

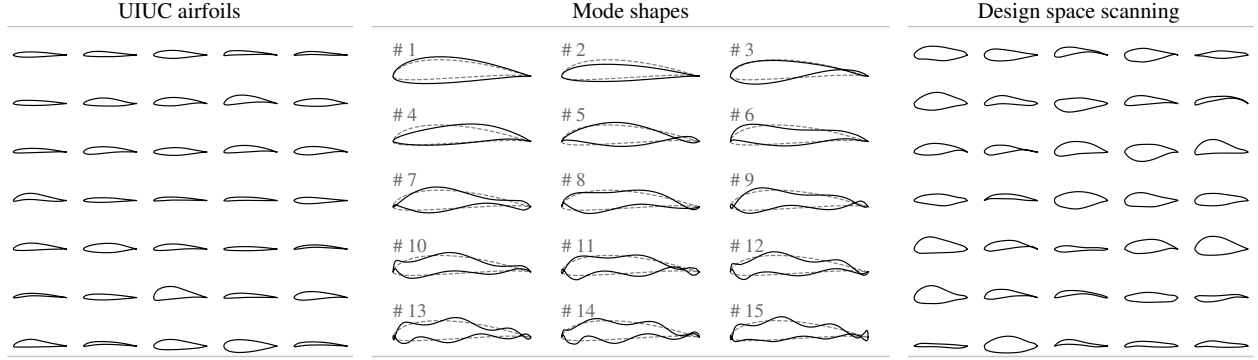


Fig. 2 Design space defined by UIUC airfoil mode shapes suffers from the issue of geometric abnormality.

B. Tailored Modal Parameterization with Deep-learning-based Geometric Filtering

To address the issue, we propose to define a compact airfoil design space by extracting modes shapes from a specific class of airfoils rather than the UIUC airfoils. These airfoils are customized based on arbitrarily specified airfoil design constraints by using the deep-learning-based optimal sampling method [24]. The optimal sampling method was developed to “traverse” possible realistic wing samples for the global wing mode shape extraction. We find this idea is also useful and even necessary for two-dimensional airfoil shape design optimization, especially in the low-Reynolds-number regime with significant laminar-to-turbulent transition phenomena. By generating samples subject to both validity and geometric constraints of the given airfoil optimization problem, the active design space suitable for the optimization problem can be compactly represented by the shape modes derived from the samples. So, in this work, we modify the optimal sampling method to generate airfoil samples for the extraction of tailored airfoil modes.

The deep-learning-based optimal sampling method is composed of two steps: 1) provide a realistic sample, 2) optimize the sample to satisfy all imposed geometric constraints. First, a generative adversarial network (GAN) [41–44] is trained using the UIUC airfoils, and the network can generate an arbitrary number of realistic airfoils. We use convolutional neural networks (CNN) as the generator and discriminator in the airfoil GAN model since it has shown to be helpful to ensure the smoothness of the synthetic airfoils [43]. The structure of the generator and discriminator are the same as those in [43]. Based on the investigation in [45], the Wasserstein distance [46] is used as the metric of distribution difference to avoid mode collapse issues of the airfoil GAN model. The synthetic airfoils generated by the GAN model usually do not meet the required geometric constraints, and thus airfoil modes directly derived from them lead to large design space as well. So, we modify the GAN airfoils using shape optimization to satisfy the constraints. Meanwhile, to maintain the sampling sparsity, the modification is performed to find the nearest feasible sample airfoil shape ($\mathbf{y}_{\text{optimal}}$) by minimizing the displacement ($d = \|\mathbf{y}_{\text{GAN}} - \mathbf{y}_{\text{optimal}}\|_2$). With different GAN airfoils as the starting points, the optimization can lead to different feasible sampling airfoils.

This idea is straightforward and an early application of similar strategies can be traced back to [47]. The difficulty is to ensure that the sample shapes are realistic. Otherwise, the sample airfoils would be abnormal and cannot represent the desired design space. To this end, we use the deep-learning-based geometric validity model developed in our previous work [45] as a constraint in the optimization. The constraint is enforced by using a discriminative model based on CNN, which is trained using labeled realistic airfoils and abnormal airfoils. The discriminative model can quickly detect the geometric abnormality without using any aerodynamic analyses, so there is almost no considerable computational cost introduced. The realistic and abnormal airfoils are generated by using the GAN model and randomly deforming the UIUC airfoils, respectively. Based on previous analyses [45], we use a CNN model with four convolutional layers trained by the MSE loss function, and the filter size (n_{size}) and the number of filters (n_{filter}) in each layer are $n_{\text{size}} = 5$ and $n_{\text{filter}} = 64$, respectively. As shown in Fig. 3, this model provides validity scores (s_{validity}) with a smooth decay from an arbitrary realistic airfoil to an abnormal airfoil, which implies a good generalization.

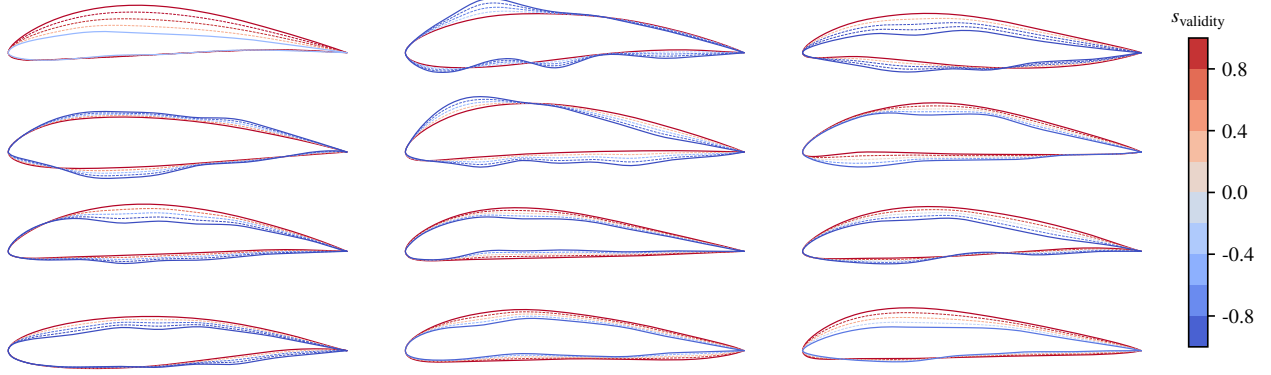


Fig. 3 Deep-learning-based geometric validity model shows a good generalization in detecting different kinds of geometric issues. Monotonic decaying of validity scores (s_{validity}) from arbitrary realistic airfoils to abnormal airfoils is valuable to geometric filtering.

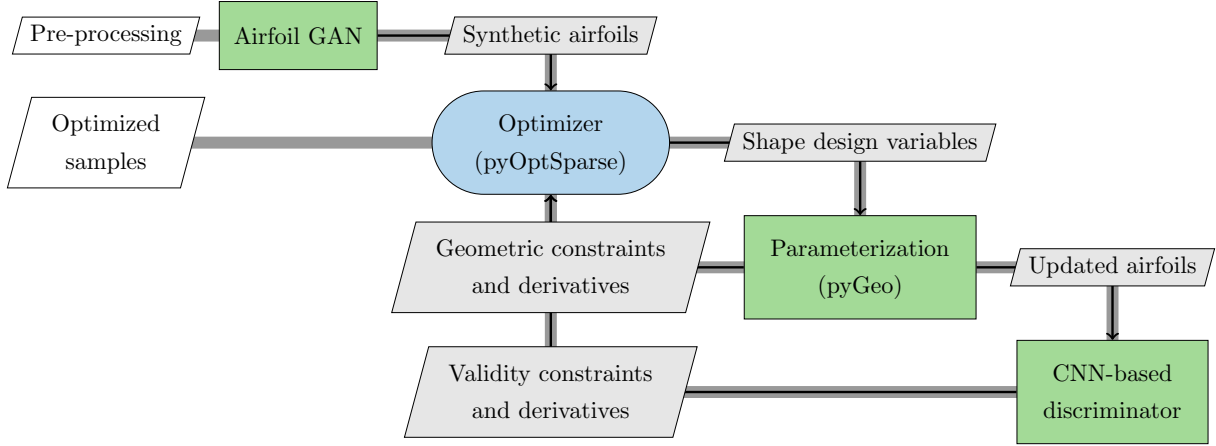


Fig. 4 Deep-learning-based optimal sampling method.

The XDSM (eXtended Design Structure Matrix) diagram [48] of the deep-learning-based optimal airfoil sampling method is shown in Fig. 4. The shape deformation is realized by using the free form deformation (FFD) method implemented in pyGeo [49]. To ensure that the sample airfoils are strictly subject to the constraints, the sequential least squares programming (SLSQP) algorithm implemented in pyOptSparse [50, 51][†] is used to solve the optimization problem.

To show the effectiveness of the proposed sampling method, some sample airfoils are shown in Fig. 5 for an airfoil design problem with geometric constraints as follows: the maximum thickness constraint ($t_{\max} = 15\%$), the thickness constraint at the 75% chord ($t_{0.75c} \geq 7\%$), and the area constraint ($A \geq 0.9 \times A_{\text{NACA0015}}$). The sampling optimization is subject to the geometric validity constraint to enforce no drop in the s_{validity} score, and the initial GAN airfoils are provided as a comparison with the final sample airfoils. As expected, geometric constraints are satisfied after the optimal shape modification and the geometric validity is maintained. Intermediate airfoil shapes during the optimization are provided to show the modification process. A complicated trade-off between the objective and constraints can be seen from the intermediate airfoils. Thus, it is vital to use the deep-learning-based validity constraint, or otherwise, the modified shape is likely to be abnormal despite the satisfaction of geometric constraints.

Despite that all geometric constraints are satisfied, some sample airfoils may not be aerodynamically feasible for the given design problem. For example, an airfoil with a relatively small leading-edge radius generally has a small

[†]<https://github.com/mdolab/pyoptsparse>

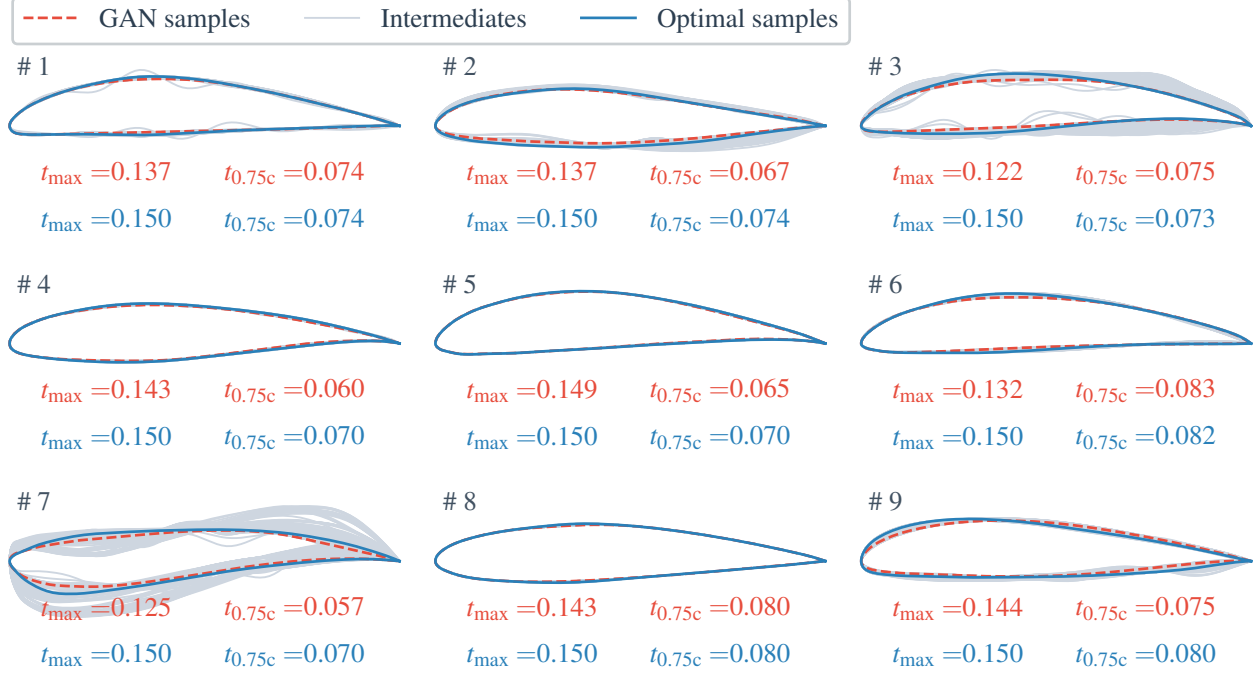


Fig. 5 Optimization process of the deep-learning-based optimal sampling method shows the complicated trade-off in finding feasible samples.

maximum lift coefficient and thus is unsuitable for high-lift airfoil design. To further shrink the design space, these sample airfoils can be excluded by performing a fast aerodynamic analysis using low-fidelity models.

With the selected samples generated by the optimal sampling method, tailored airfoil mode shapes can be derived using Eq. 2. As shown in Fig. 6, the airfoil modes for airfoil design subject to three sets of geometric constraints are provided. For each set of geometric constraints, 500 sample airfoils are generated by the proposed sampling method, and $s_{\text{validity}} \geq 0.75$ is used to leave enough room for geometric innovation [45]. Before airfoil mode derivation, the samples are further filtered by a fast analysis with Xfoil [52] at $Re = 5.0 \times 10^5$ (Reynolds number) and $M = 0.1$ (Mach number) to ensure $C_l^{\max} \geq 1.5$. Design space scanning is performed using randomly generated sample airfoils via LHS, which do not encounter any serious geometric abnormality issues compared with those in Fig. 2. This implies that the design space spanned by the tailored airfoil modes is much more compact and it is beneficial to improve optimization efficiency.

C. Optimization Framework

We use the efficient global optimization (EGO) method [14] implemented in pyAeroEGO [24, 43] to perform airfoil shape design. EGO is composed of two phases: the design of the experiment (DoE) process and the infill process. In DoE, the initial sample airfoils are generated by LHS in the design space spanned by the tailored airfoils modes. The kriging model implemented in the Surrogate Modeling Toolbox (SMT) [53] [‡] is used as surrogate models of aerodynamic coefficients. The infill process is performed iteratively, and, in each iteration, sub-optimization problems are solved, which consist of a minimization of the objective function and a maximization of the expected improvement (EI) function. Geometric constraints are enforced in solving both sub-optimization problems. To strictly handle the constraints, we use the SLSQP algorithm to solve sub-optimization problems. A multi-start strategy is adopted to reduce the possibility of being stuck at local minima, which is more effective than gradient-free algorithms in aerodynamic design problems [54].

Xfoil [52] is used as the aerodynamic solver in this work based on two reasons. First, the accuracy of Xfoil is acceptable in the subsonic aerodynamic analysis of low-Reynolds-number airfoils. Xfoil uses a linear-vorticity panel method coupled with an integral boundary-layer method for viscous analysis and adopts the e^N method [55] to evaluate the transition position. The suitability of Xfoil for low-Reynolds-number airfoil design has been demonstrated in the

[‡]<https://github.com/SMTorg/smt>

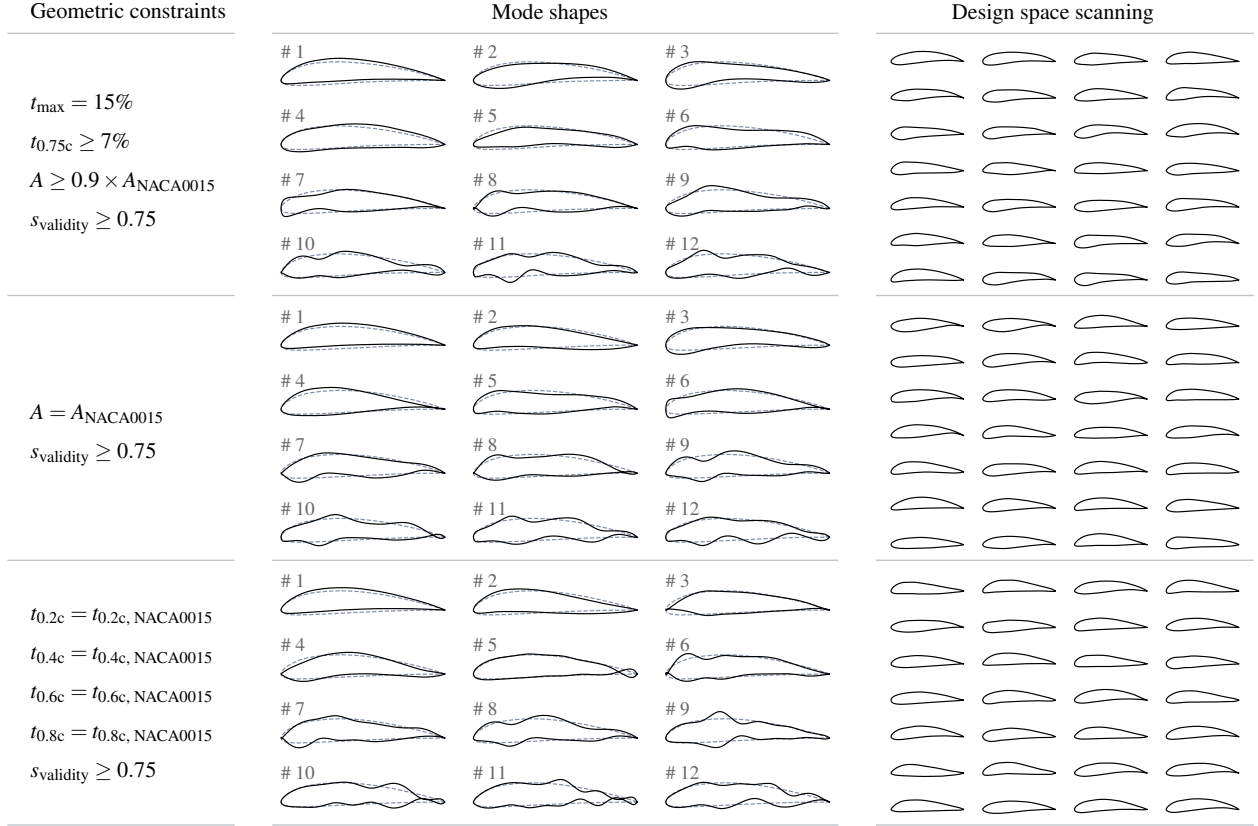


Fig. 6 The design space defined by the tailored mode shapes is much more compact and does not encounter the issue of geometric abnormality.

literature [28, 33, 56]. For example, Ma et al. [28] showed that Xfoil predictions were in good agreement with wind tunnel experiment data [57] in the analysis of the E387 airfoil at $Re = 4.6 \times 10^5$; Morgado et al. [56] showed that Xfoil provided comparable aerodynamic analysis results to those obtained by solving Reynolds-averaged Navier-Stokes (RANS) equations with a transition model and wind tunnel experiments. The accurate capability was held for different airfoil shapes at the considered Reynolds numbers (2.0×10^5 and 4.7×10^5); Leloudas et al. [33] showed that Xfoil provided very similar results to the UIUC low-turbulence subsonic wind tunnel experiments [58] in the analysis of the RG15 airfoil at $Re = 6.0 \times 10^4$ and $Re = 3.0 \times 10^5$. It was also shown that RANS without a suitable transition model would over-estimate the drag coefficients. Second, the low computational cost of aerodynamic analyses by Xfoil makes it computational acceptable to comprehensively validate the proposed method in airfoil design via multiple optimization tests. Since Xfoil is used in a black-box way in our optimization framework, it is a good demonstration of the portability, and airfoil design optimization using any other solvers such as commercial software and legacy in-house codes can also benefit from the proposed method.

To avoid communicating with the Xfoil executable by reading/writing files, a stripped-down version of Xfoil in the form of a Python module [§] is used, which is convenient to be coupled with our Python-based optimization framework. The XDMS diagram of the optimization framework is shown in Fig. 7.

III. Optimization Efficiency

To validate the proposed method, optimization efficiency using conventional airfoil modes and tailored airfoil modes is compared in this section. The investigated airfoil optimization is to minimize the drag coefficient C_d at $Re = 5.0 \times 10^5$ and $M = 0.1$ with a cruise lift coefficient of $C_l = 1.1$. Other constraints on the maximum lift coefficient, the pitching moment, thicknesses, and the airfoil area are shown in Table 1. Fifteen airfoil modes are used as the shape design

[§]<https://pypi.org/project/xfoil>

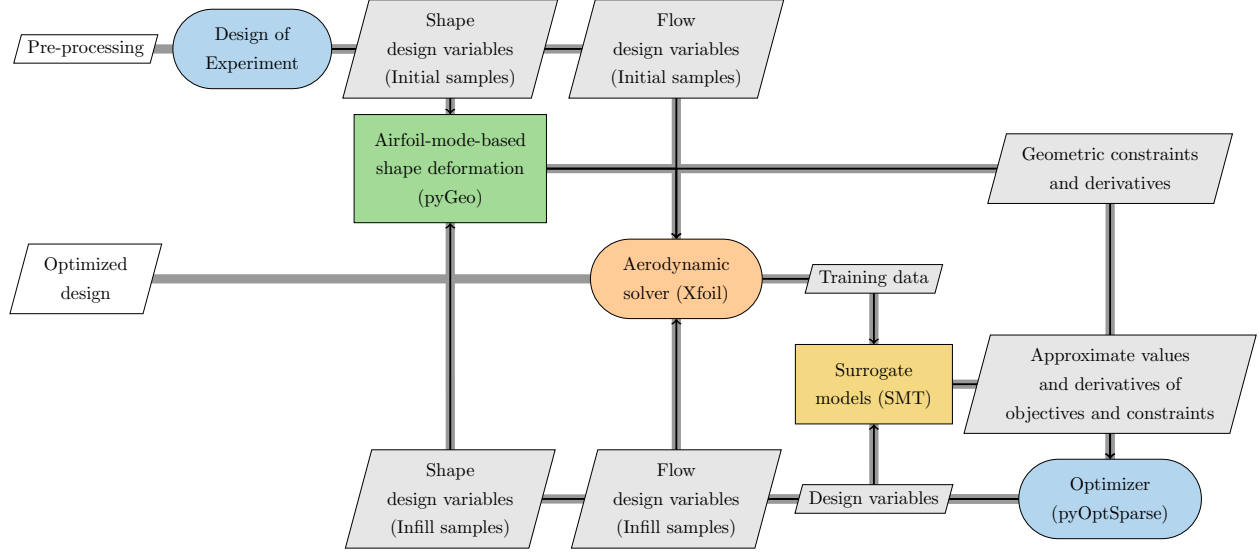


Fig. 7 XDSM diagram of the surrogate-based airfoil design optimization using tailored airfoil modes.

variables. The cruise angle of attack is an independent design variable to be optimized together with the airfoil shape.

Table 1 Statement of airfoil design optimization for drag minimization.

	Functions	Description	Quantity
Minimize	C_d	Drag coefficient	1
With respect to	$\mathbf{x}_{\text{shape}}$	Coefficients of airfoil modes	15
	α	Angle of attack	1
		Total design variables	16
Subject to	$C_l = 1.1$	Lift constraint	1
	$C_l^{\max} \geq 1.85$	Stall constraint	1
	$C_m \geq -0.15$	Quarter-chord pitching moment constraint	1
	$t_{\max} = 15\%$	Maximum thickness constraint	1
	$t_{0.75c} \geq 7\%$	Thickness constraint at 75% chord	1
	$A \geq 0.9 \times A_{\text{NACA0015}}$	Area constraint	1
		Total constraints	6

Four surrogate models (for C_d , C_l , C_m , and $C_l^{\max}$) are constructed in this case. The first three models are functions of both shape design variables and the angle of attack, while the last one for $C_l^{\max}$ is merely dependent on shape design variables. $C_l^{\max}$ is achieved at a high angle of attack just before the stall. We solve the lift coefficients in a large range of angles of attack from 10° to 22° with an incremental of 0.25° , and then the largest coefficient among them is used to approximate $C_l^{\max}$ of the examined airfoil shape.

The convergence histories of optimization using conventional and tailored modes are shown in Fig. 8. One hundred initial samples are generated in the DoE process, and multiple tests are performed to reduce the influence of the initial samples. The shaded regions in the figure show the convergence of different tests, and the solid/dash lines are the averaged convergence curves. Airfoil design optimization using conventional airfoil modes is inefficient, and multiple tests lead to distinct results, which implies non-convergence of the optimization. Using tailored airfoil modes significantly improves the optimization efficiency, and multiple optimization tests merely differ in an early period and eventually converge to the same result.

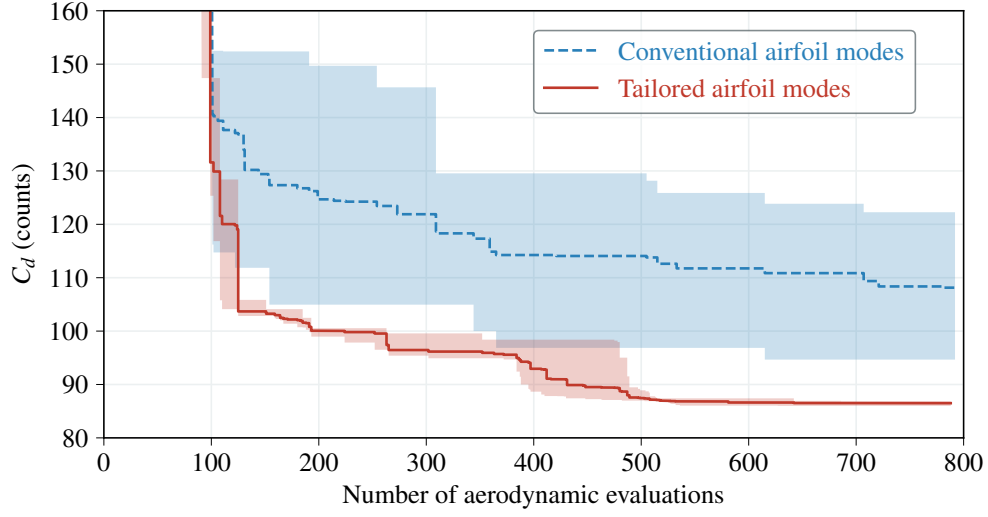


Fig. 8 Convergence comparison of airfoil design optimization using conventional and tailored modes.

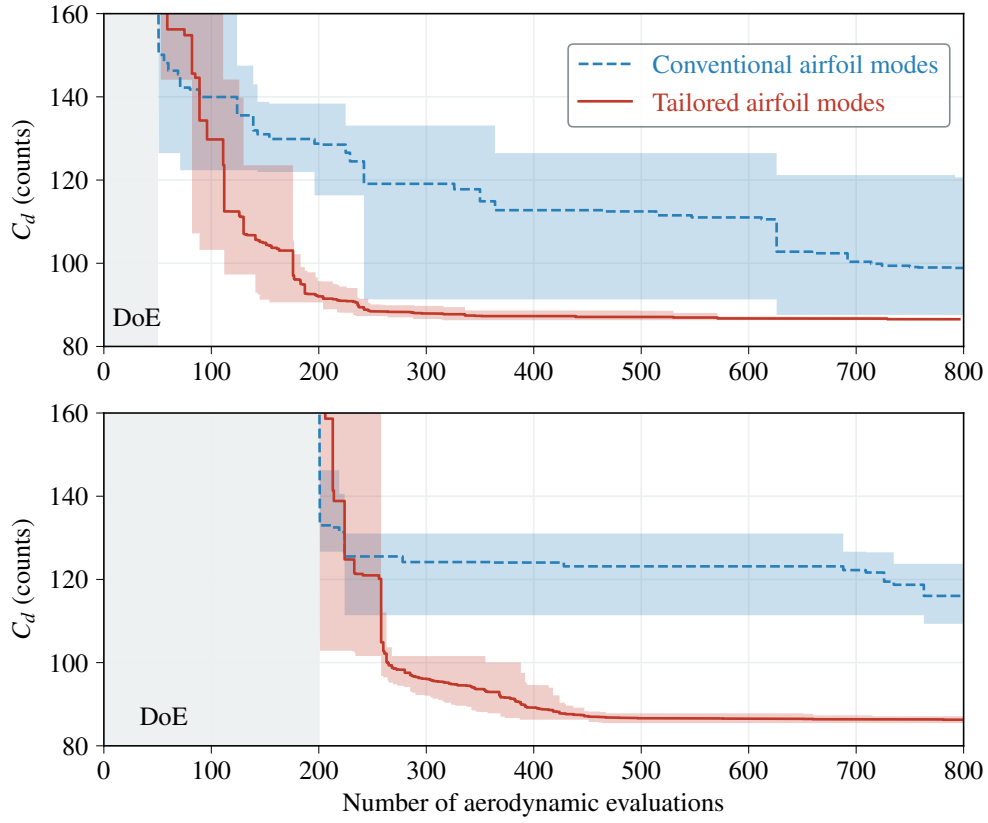


Fig. 9 Convergence comparison of airfoil design optimization using conventional and tailored modes.

Optimization efficiency may differ when the optimization starts from a different number of initial DoE samples. To exclude this influence, we perform airfoil design optimization using another two numbers of DoE samples: 50 and 200. Multiple tests are involved as well to have a fair comparison on the averaged convergence curves. As shown in Fig. 9, no matter starting with how many DoE samples, optimization using the tailored airfoil modes is always much more

efficient. Airfoil design using tailored airfoil modes in different tests converges to almost the same drag value, which shows the reliable convergence once again.

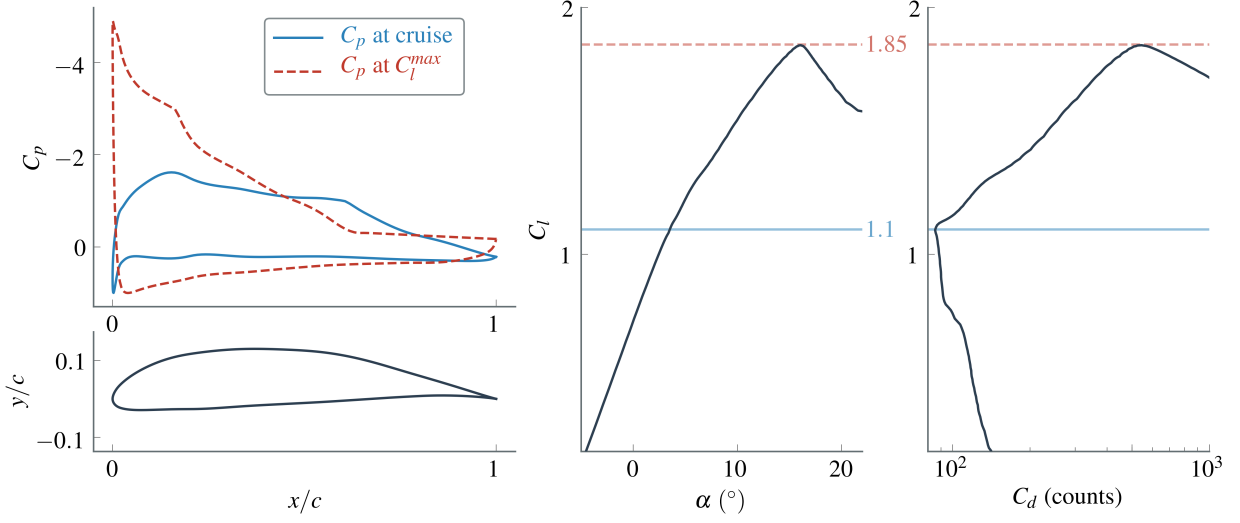


Fig. 10 Optimized airfoil using tailored modes subject to $C_l^{\max} \geq 1.85$.

The optimized airfoil shape using the tailored modes is shown in Fig. 10. The transition position on the upper surface of the optimized airfoil is at $x_{tr} \approx 0.61$, and the C_p slope ahead of this point is small to maintain a laminar flow for drag reduction. The maximum lift is achieved at $\alpha = 16.0^\circ$. At the adjacent domain of this angle, we find that the polar curve is not smooth, which means that there is an uncertainty in evaluating $C_l^{\max}$. So, compared with aerodynamic functions at cruise, it is more difficult for a surrogate model to accurately predict $C_l^{\max}$. However, such a constraint is usually required in practical design. The choice of $C_l^{\max}$ constraint may influence optimization efficiency. We further investigate optimization efficiency subject to a larger $C_l^{\max}$ constraint ($C_l^{\max} = 2.1$). As shown in Fig. 11, the advantage of tailored airfoil modes is still significant. The difference in the averaged optimized C_d values by conventional airfoil modes and tailored airfoil modes is over 20 counts.

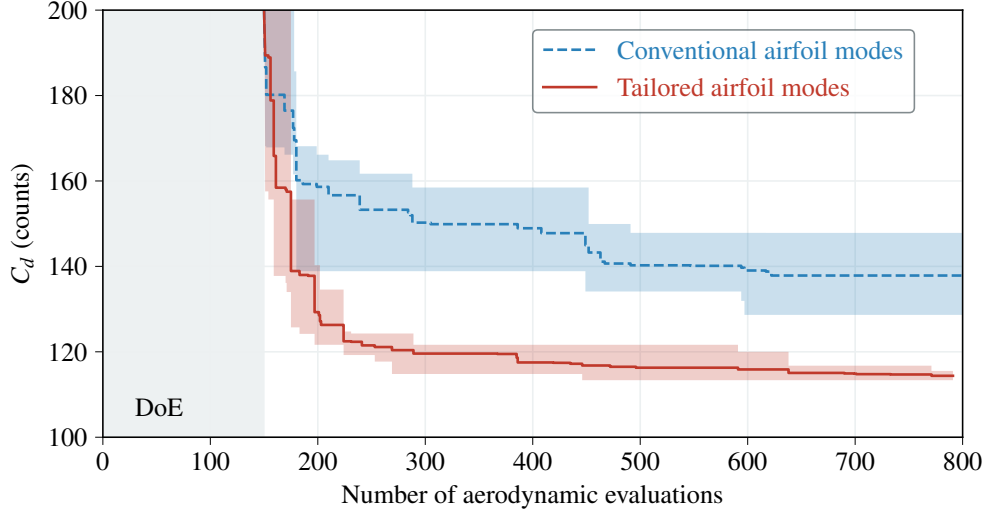


Fig. 11 Convergence comparison of airfoil design optimization using conventional and tailored modes subject to $C_l^{\max} \geq 2.1$.

The optimized airfoil with $C_l^{\max} = 2.1$ using the tailored airfoil modes is shown in Fig. 12. Compared with the

optimized airfoil subject to $C_l^{\max} = 1.85$, the maximum camber position of this airfoil is moved ahead to achieve a higher $C_l^{\max}$, and correspondingly the transition position at cruise is changed to $x_{tr} = 0.38$, which leads to a larger C_d at $C_l = 1.1$. The lift or polar curve around $C_l^{\max}$ is associated with more significant non-smoothness. Nevertheless, optimization using the tailored airfoil modes still shows high efficiency, which means that the proposed method is practical for airfoil design with challenging aerodynamic constraints.

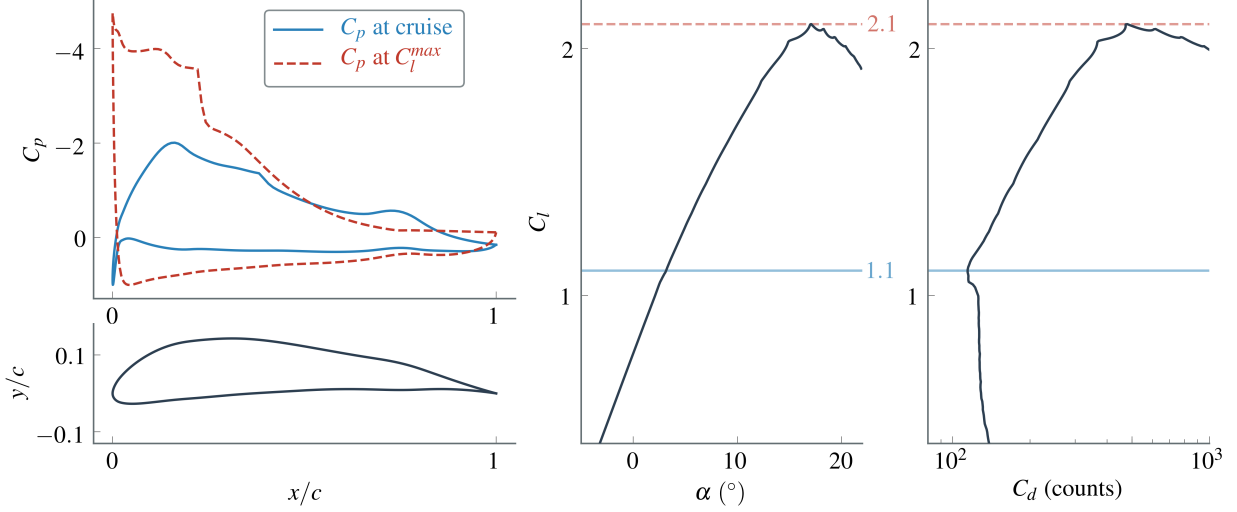


Fig. 12 Optimized airfoil using tailored modes subject to $C_l^{\max} \geq 2.1$.

Overall, after performing airfoil design optimization in various cases considering influences from different aspects, it has been shown that optimization efficiency by using tailored airfoil modes is much higher than conventional modes. Since a compact design space fundamentally reduces the optimization difficulty, design optimization using other methods (for example those based on neural-network-based aerodynamic analysis models) can also benefit from this contribution. In the next section, we will show the application of the proposed method in practical airfoil design.

IV. Practical Design of Low-Reynolds-number High-Lift Airfoils

Optimization efficiency of using tailored airfoil modes in single-point low-Reynolds-number airfoil design has been validated. However, designs solved by single-point optimization may be impractical, which has been demonstrated in previous studies [4, 5]. So, we perform multi-point design optimization of the low-Reynolds-number high-lift airfoil (Sec. 1) and investigate the influence of $C_l^{\max}$ constraint on the optimal cruise C_l in low-Reynolds-number UAV airfoil design (Sec. 2).

A. Multi-point Design Optimization

To investigate the robustness of the optimized airfoil solved by single-point design (Fig. 10), we compute the drag coefficients in the adjacent C_l - Re domain of the cruise point ($Re = 5.0 \times 10^5$ and $C_l = 1.1$). As shown in Fig. 13, the single-point design leads to an abrupt increase of C_d with the increase of C_l when $C_l \geq 1.2$. This means that the aerodynamic performance of the single-point design is very sensitive to the perturbation of the angle of attack and thus it would be unsuitable for practical applications.

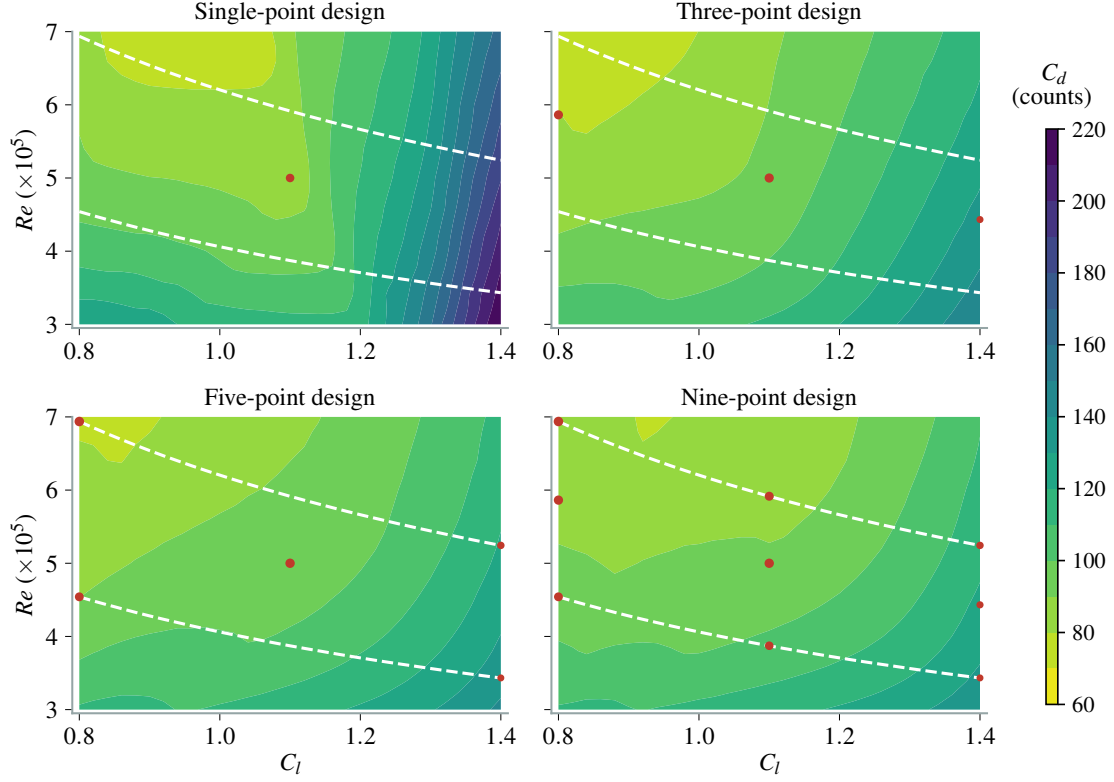


Fig. 13 C_d contours of optimized airfoils solved by single-point design and multi-point design. The red dots show the design points with their sizes indicating the combination weights. The two white dash lines are determined by considering a $\pm 40\%$ change of $C_l \times Re^2$.

We use multi-point design optimization to improve the robustness of the optimized airfoil to the change of Re and C_l . For a UAV with a fixed weight, the required lift is $L = C_L \times 0.5\rho U^2 S$, where U and S are the flight speed and reference area of the UAV, respectively. Neglecting the influence of density and viscosity changes, Re scales linearly with U . So, the lift is approximately proportional to the parameter $C_l \times Re^2$ in airfoil design. To account for the change of fuel or payload in practical applications, we evaluate the overall aerodynamic performance with $\pm 40\%$ change of $C_l \times Re^2$ relative to the single-design point. The white dash lines in Fig. 13 show the bounds of the concerned C_l-Re domain. Three sets of multi-point optimization are performed, and the design points are shown in Fig. 13. The objective function of multi-point design is the weighted average of C_d . We chose the combination weights ($w_i, i = 1, \dots, n$) of different design points approximately uniformly by referring to the C_d values in single-point design, that is,

$$w_i = \frac{1/C_d^i}{\sum_{j=1}^n 1/C_d^j}, \quad (3)$$

where C_d^i is the drag coefficient at the i th design point of the optimized airfoil solved by single-point optimization. This choice could prevent the multi-point optimization from focusing too much on the design points with large drag coefficients such as those with large C_l and small Re in this case.

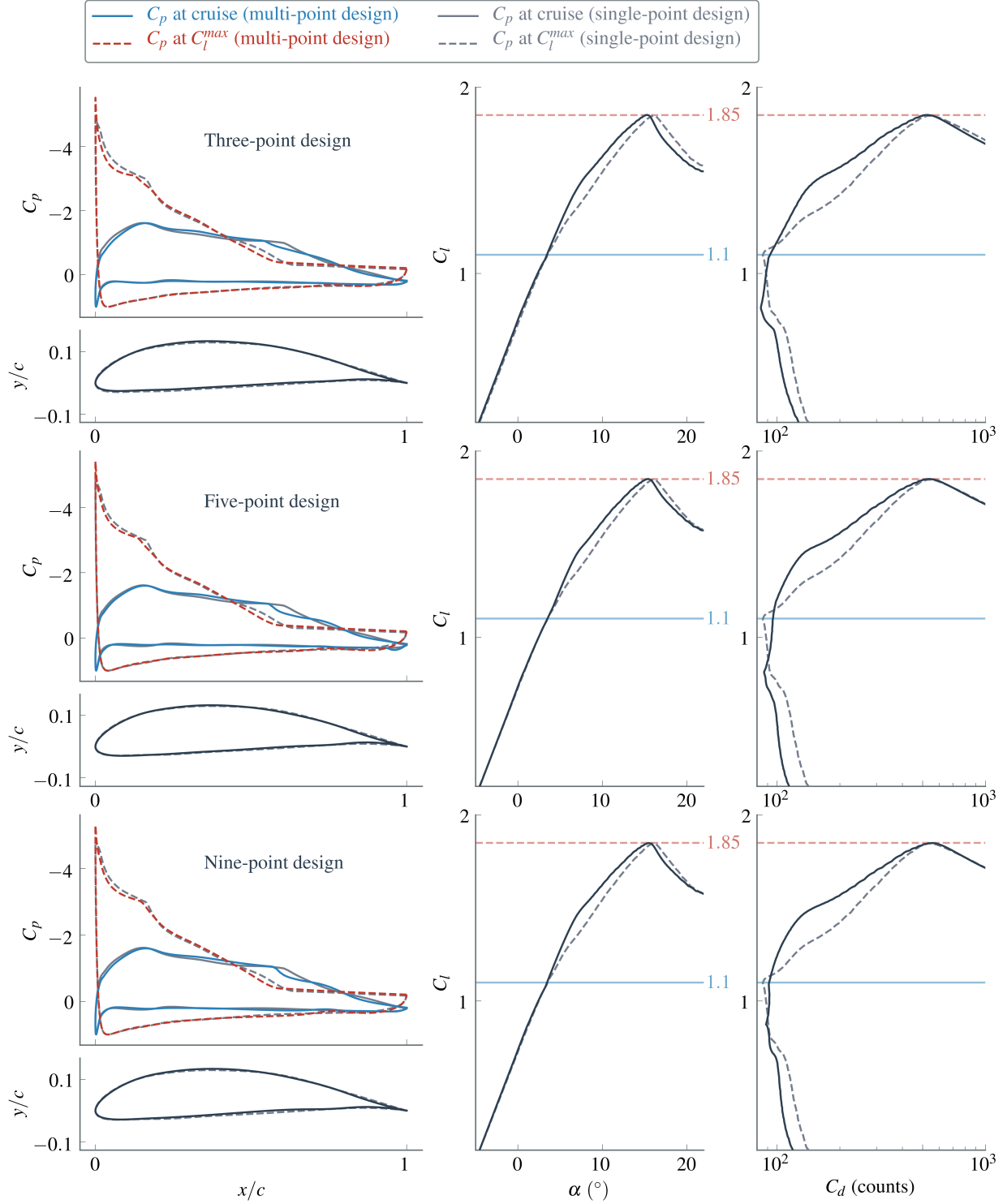


Fig. 14 Comparison of optimized airfoils solved by single-point and multi-point design. The lift curve and drag polar curve are evaluated at $Re = 5.0 \times 10^5$.

The drag contours of optimized airfoils solved by multi-point design are shown in Fig. 13. The abrupt drag increase issue in the single-point design when $C_l > 1.2$ is addressed in all multi-point design cases. As a consequence, the area of the contour of $C_d \leq 100$ counts increases significantly. The optimized airfoils are shown in Fig. 14, and the polar

Table 2 Summary of the optimized airfoils.

	C_d^{mean}	C_d^{cruise}	C_l^{max}	C_m^{cruise}	t_{max}	$t_{0.75c}$
Single-point design	108.3	85.5	1.85	-0.150	15.0%	7.0%
Three-point design	98.2	92.4	1.85	-0.150	15.0%	7.0%
Five-point design	98.8	95.4	1.85	-0.150	15.0%	7.0%
Nine-point design	97.0	91.5	1.85	-0.150	15.0%	7.0%

curves at $Re = 5.0 \times 10^5$ are compared. The single-point design result is provided as a comparison. There is merely a minor difference between the optimized airfoil shapes obtained by single-point design and multi-point design. However, the difference in lift and drag polar curves is significant. The minor shape modification by multi-point design changes the transition position on the upper airfoil surface, which makes the laminar-to-turbulent transition less sensitive to the change of Re and the angle of attack. Consequently, multi-point designs maintain small drag coefficients in the adjacent $C_l - Re$ domain, and the robustness to the change of angle of attack is also improved.

The optimization results are summarized in Table 2, where the average drag C_d^{mean} is evaluated by 321 points distributed in the concerned $C_l \times Re^2$ domain shown in Fig. 13. The specified constraints are satisfied in all design optimizations. Compared to the single-point design, any of the three multi-point design optimizations significantly reduces the average drag. Increasing the number of design points and defining their combination weights based on a mission frequency analysis [59] may further improve the performance. Nevertheless, the three-point design optimization is shown to be effective in improving the design robustness.

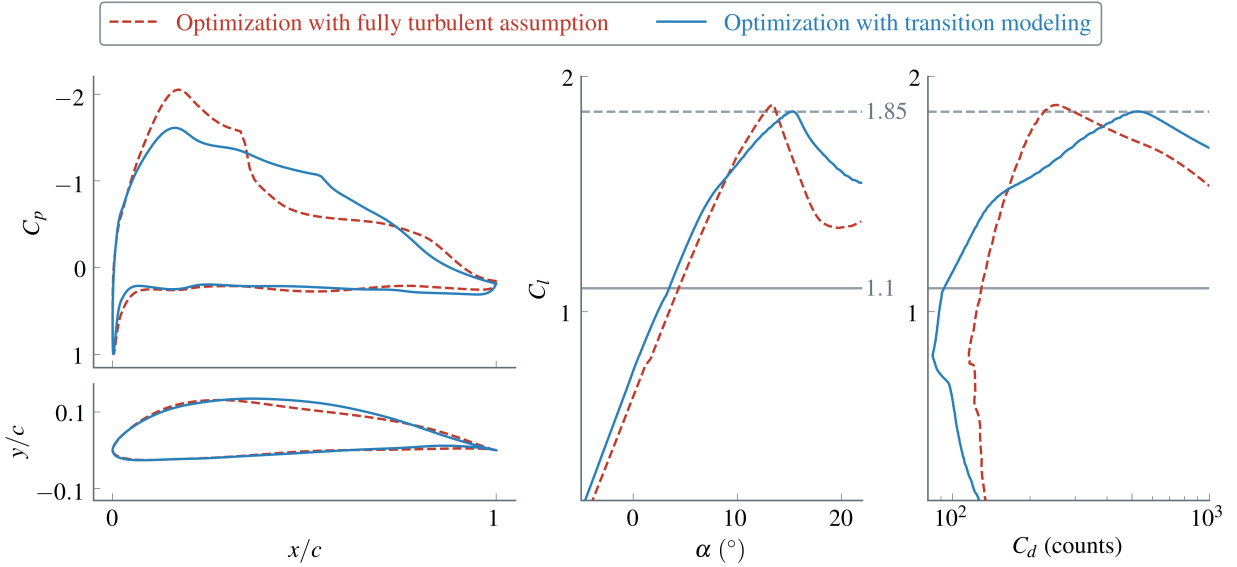


Fig. 15 Optimization based on aerodynamic analyses with the fully turbulent assumption cannot provide a reasonable design since the optimization benefit in low-Reynolds-number airfoil design is mainly achieved by pushing the flow transition point backward.

In previous investigations, we find that the optimization trade-off is mainly surrounding the laminar-to-turbulent transition point on the upper surface. It implies that optimization based on the fully turbulent assumption would not provide a reliable design in the low-Reynolds-number regime as warned by previous researches such as [60]. To further verify this point, we perform another three-point optimization of the airfoil with the cruise performance evaluated with the fully turbulent assumption, which is achieved by fixing the transition points to the leading edge in Xfoil. The optimized airfoils are compared in Fig. 15, where the performance is evaluated by aerodynamic analyses with modeling the laminar-to-turbulent transition. With the fully turbulent assumption, the optimized airfoil has the maximal thickness position closer to the leading edge, and this leads to a lower pressure at the suction peak. Airfoil thickness

after the maximal thickness point reduces significantly and thus leads to a larger adverse pressure gradient that induces laminar-to-turbulent transition. This disadvantageous effect cannot be detected in aerodynamic analyses subject to the fully turbulent assumption. On the contrary, optimization with the transition modeling prevents this negative effect, and the optimized airfoil has a flatter upper surface to keep the flow laminar as far as possible. Thus, it is necessary to model the laminar-to-turbulent transition in low-Reynolds-number airfoil design. Otherwise, it can be anticipated that the optimized results would not be practical even using high-fidelity RANS with a turbulence model.

B. Optimal Cruise C_l subject to a Practical $C_l^{\max}$ Constraint

In UAV airfoil design, the maximal lift coefficient ($C_l^{\max}$) is a critical factor for taking-off, landing, and preventing stall for flight safety [61]. To leave a safety margin, a choice of the industry is ensuring $C_l^{\max} \geq C_l^{\text{cruise}}/0.56$, where C_l^{cruise} is the designed lift coefficient at cruise. Although it could be beneficial to increase the lift-drag ratio L/D or the endurance factor $L^{1.5}/D$ by flying at a large C_l^{cruise} , the constraint on $C_l^{\max}$ poses an additional restriction on the selection of the optimal C_l^{cruise} . To investigate this effect, we perform airfoil design optimization at different C_l^{cruise} subject to the $C_l^{\max}$ constraint.

Based on the investigation in the last section, we perform multi-point design optimization to improve the design robustness, and three design points ($C_l = C_l^{\text{cruise}} - 0.3, C_l^{\text{cruise}}, C_l^{\text{cruise}} + 0.3$) are chosen from the $C_l \times Re^2$ contour line passing the cruise point. The weights of different design points are defined using Eq. 3. Other constraints on thicknesses, area, and pitching moment are the same as specified in Table 1.

The optimized airfoils, lift curves, polar curves, and the design points are shown in Fig. 16. The constraint on $C_l^{\max}$ is always active. With the increase of C_l^{cruise} , a significant effect on the optimized airfoil is the increase of the airfoil camber, and the transition position on the upper surface (indicated by red dots in Fig. 16) moves forward slightly when $C_l^{\text{cruise}} \leq 1.05$. When $C_l^{\text{cruise}} > 1.05$, the maximum camber positions of optimized airfoils are moved forward to generate a larger $C_l^{\max}$. In this circumstance, a slight increase of C_l^{cruise} leads to significant moving-forward of the transition position and thus produces a much larger C_d . Thus, there is an active trade-off between the transition position at cruise and $C_l^{\max}$ in the practical airfoil design.

Commonly used objective functions in UAV airfoil design are L/D and $L^{1.5}/D$, and both can be transformed to a series of drag minimization problems at different C_l^{cruise} . The two objective functions of the optimized airfoils at different C_l^{cruise} are shown in Fig. 17. It can be seen, subject to the practical constraints introduced above, the optimal cruise lift to maximize the objective functions is about 1.05. We find that this result holds in another airfoil design problem with different geometric constraints ($t_{\max} = 18\%$ and $t_{0.75c} \geq 7\%$). We are working on the validation of the design by conducting wind tunnel experiments, and the results will be reported in future work. This finding may be guidance for similar airfoil design problems. Nevertheless, the optimal C_l^{cruise} may differ significantly if the constraint on $C_l^{\max}$ is changed. So, it is still highly recommended to independently perform airfoil shape design optimization for specific application problems. This translates to lots of work needed and thus it is of great significance to use the tailored airfoil modes to increase optimization efficiency.

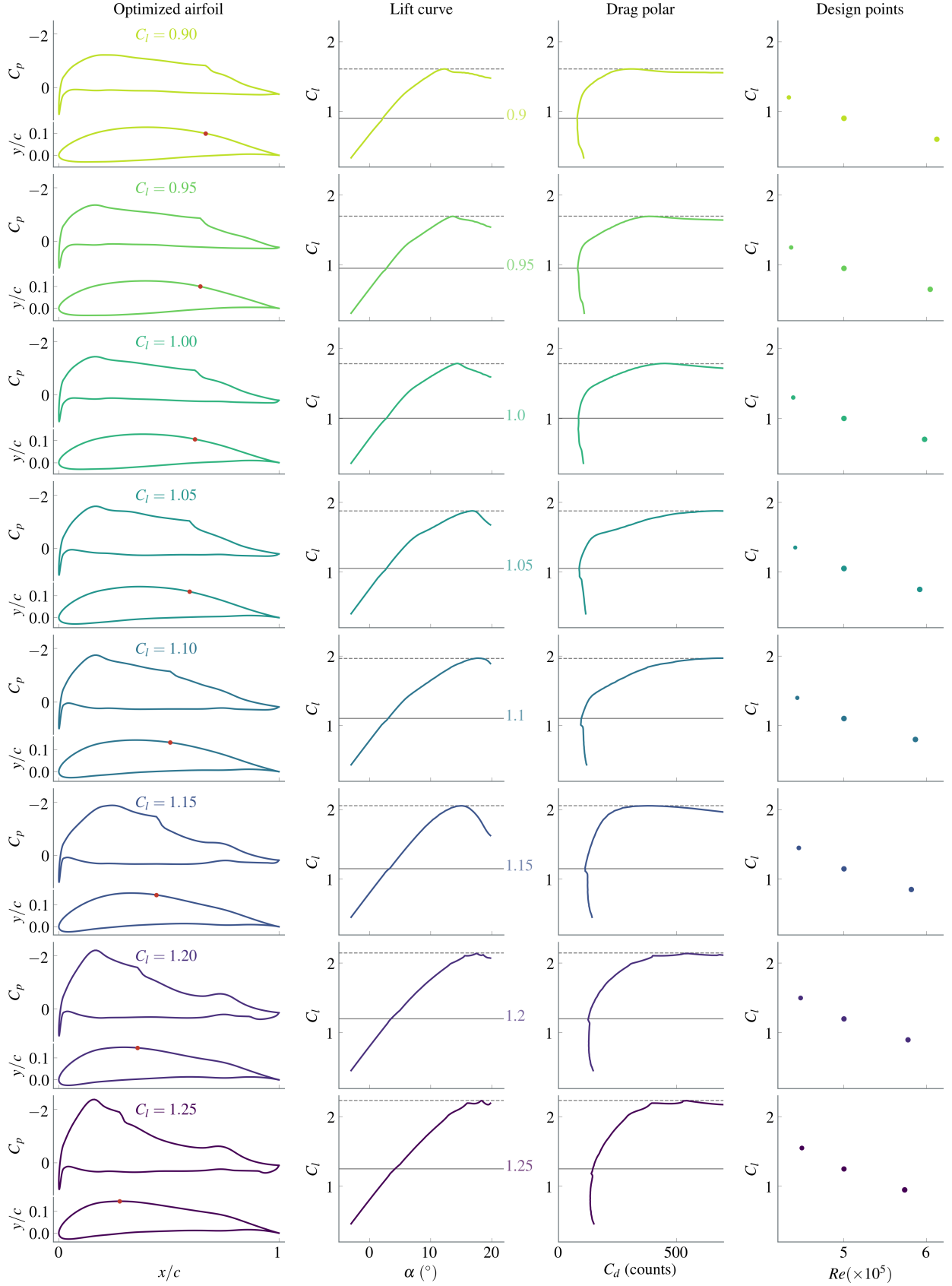


Fig. 16 Optimized airfoils for different C_l^{cruise} subject to $C_l^{\text{max}} \geq C_l^{\text{cruise}}/0.56$. Red dots on airfoil shapes indicate the position of flow transition on the upper surface. Lift and drag polar curves are evaluated at $Re = 5.0 \times 10^5$.

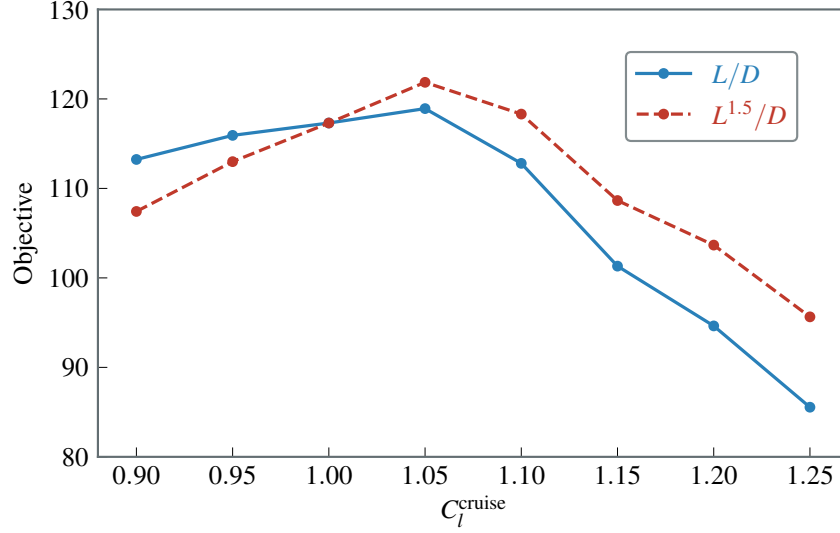


Fig. 17 L/D and $L^{1.5}/D$ of optimized airfoils solved at different cruise C_l .

V. Conclusions

This work presents a tailored airfoil modal parameterization method to improve optimization efficiency in low-Reynolds-number airfoil design. The tailored airfoil modes are extracted from customized airfoils generated by the deep-learning-based optimal sampling method based on the required geometric constraints. Satisfying both the deep-learning-based validity constraint and geometric constraints, these sample airfoils represent the feasible domain for the airfoil shape design problem. Compared with conventional airfoil modes derived from a general airfoil database, the tailored airfoil modes significantly improve the efficiency of surrogate-based optimization in low-Reynolds-number airfoil design.

Using the proposed method, we find that the effect of laminar-to-turbulent transition has to be considered in low-Reynolds-number airfoil design. Multi-point optimization is necessary to obtain a robust design. Otherwise, the optimized airfoil is impractical since the transition position is sensitive to the change of the angle of attack or the Reynolds number. There is a trade-off between the transition position at cruise and the maximal lift coefficient (C_l^{max}) before stall, and thus airfoil shape design optimization merely considering the cruise performance would reduce C_l^{max} . In the investigated cases, $C_l^{\text{max}} \geq C_l^{\text{cruise}}/0.56$ is always an active constraint, and the optimal cruise lift for maximizing L/D and $L^{1.5}/D$ is $C_l^{\text{cruise}} \approx 1.05$.

These results show that low-Reynolds-number high-lift airfoil design is a complex task where multiple design points and constraints have to be reasonably considered. Thus, with a significant improvement of optimization efficiency, the proposed method is of great importance to low-Reynolds-number airfoil design and UAV design.

Acknowledgments

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